

Experiment 7

7.1.2 Possible combinations from 3 digits

Algorithm

Step 1: Start the program.

Step 2: Read three space-separated digits as input (a, b, c).

Step 3: Store the three digits in a list or separate variables.

Step 4: Print the first combination by arranging digits as a, b, c.

Step 5: Print the second combination by arranging digits as a, c, b.

Step 6: Print the third combination by arranging digits as b, a, c.

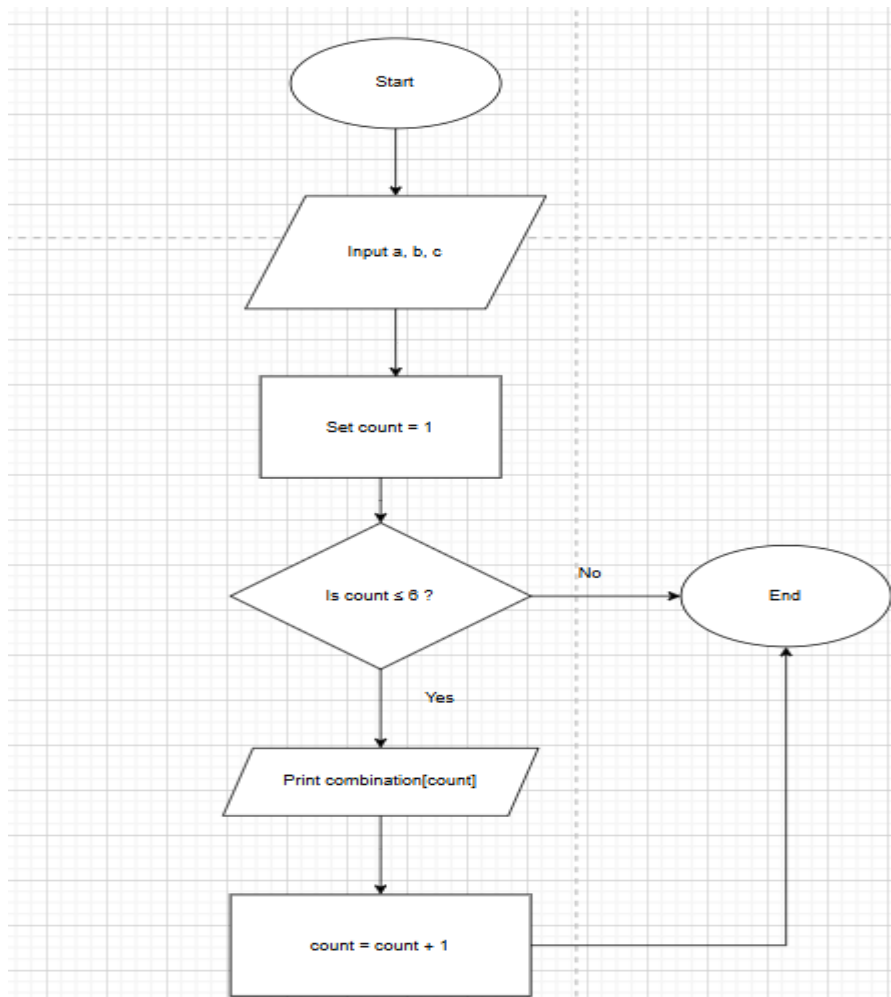
Step 7: Print the fourth combination by arranging digits as b, c, a.

Step 8: Print the fifth combination by arranging digits as c, a, b.

Step 9: Print the sixth combination by arranging digits as c, b, a.

Step 10: Stop the program.

Flowchart



Experiment 7

PYTHON CODE

```
a, b, c = input().split()
digits = [a, b, c]
for i in range(3):
    for j in range(3):
        for k in range(3):
            if i != j and j != k and i != k:
                print(digits[i] + digits[j] + digits[k])
```

Execution

The screenshot displays the CodeTANTRA online IDE interface. On the left, the problem statement for '7.1.2. Possible Combinations from Three Digits' is shown, including input/output formats and constraints. The main editor contains the Python code for generating permutations of three digits. The right sidebar shows the execution results, indicating that 3 out of 3 shown test cases and 4 out of 4 hidden test cases passed. A table compares the expected output with the actual output for the first test case.

Problem Statement: 7.1.2. Possible Combinations from Three Digits

Write a Python program to accept three digits and print all possible 3-digit combinations that can be formed using those three digits. Each digit should be used exactly once in each combination.

For three digits a , b , and c , there are 6 possible arrangements based on their indices: abc , acb , bac , bca , cab , cba .

Input Format:

- Single line contains three space-separated non-negative integers, a , b , and c , representing the three digits

Output Format:

- Print all possible 3-digit combinations
- Each combination should be printed as a 3-digit number on a separate line
- Print combinations as they appear (maintain leading zeros if first digit is 0)

Constraints:

- $0 \leq \text{each digit} \leq 9$
- Digits can be same or different

Sample Test Cases

Execution Results:

- Average time: 0.003 s, Maximum time: 0.005 s
- 3 out of 3 shown test case(s) passed
- 4 out of 4 hidden test case(s) passed

Test case 1:

Expected output	Actual output
1 2 3	1 2 3
123	123
132	132
213	213
231	231
312	312